4-5-1: Introduction to CIDR

At the end of this episode, I will be able to:

1. Identify and explain the purpose of Classless Inter-Domain Routing or CIDR in networks, given a scenario.

Learner Objective: *Identify and explain the purpose of Classless Inter-Domain Routing or CIDR in networks, given a scenario*

Description: In this episode, the learner will examine Classless Inter-Domain Routing or CIDR. We will explore the benefits that CIDR provides for IPv4 addressing and the Internet.

- What is CIDR (Classless Inter-Domain Routing)?
    - A method for allocating IP addresses and managing IP routing.
    - Introduced to improve the efficiency and scalability of IP address allocation and routing on the Internet.
- Was there any other reasons for introducing CIDR?
    - To counter the rapid exhaustion of IPv4 addresses
    - To decrease the size and complexity of routing tables on the Internet.
- What are some key features of CIDR?
    - Classless system
        - CIDR allows for a more flexible allocation of IP addresses.
        - Doesn't limit networks to predefined sizes.
    - Slash notation:
        - Introduces slash notation (example: 192.168.1.0/24)
            - This defines the network prefix and the length of this prefix
            - The number after the slash indicates the number of bits used for the network portion.
    - Efficient routing
        - By aggregating multiple IP addresses into a single entry in routing tables
        - Reduces the size of these tables, leading to more efficient and faster IP routing.
- What has the impact on Internet been since CIDR was introduced?
    - Routing table aggregation
        - The ability to aggregate routes into fewer entries (reducing the size of the routing tables on the Internet)
        - Has helped to manage the growth of routing tables (crucial for internet scalability)

- Additional Resources
    - Route summarization
        - Individual networks - 35.4.0.0/16, 35.5.0.0/16
        - Summarized route - 35.4.0.0/15
    - Route summarization
        - Individual networks - 192.168.20.0/24, 192.168.21.0/24, 192.168.22.0/24
        - Summarized route - 192.168.20.0/22
    - Route summarization
        - Individual networks - 10.10.0.0/24, 10.10.1.0/24, 10.10.2.0/24, 10.10.3.0/24, 10.10.4.0/24
        - Summarized route - 10.10.0.0/21